

Pritam

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TECHNICAL SKILLS

Programming Languages: C/C++, Python, Java, SQL, VHDL

Technologies & Frameworks: Spring Boot, Java Swing, AWT, OpenCV, REST APIs, Multithreading

Machine Learning: Linear Regression, Logistic Regression, Random Forest, Scikit-learn

Tools & Platforms: Git, GitHub, Linux, IntelliJ IDEA, VS Code, Vim

Specializations: Artificial Intelligence, Machine Learning, Robotics, Embedded Systems, PCB Design

EDUCATION

Deenbandhu Chhotu Ram University of Science and Technology

Bachelor of Technology in Electronics & Communication Engineering; CGPA: 8.5/10

Sonipat, Haryana

Sep 2023 - Present

Central Board of Secondary Education (CBSE)

Higher Secondary Certificate - Mathematics, Physics, Chemistry; Grade: 96.2%

2022

New Delhi

EXPERIENCE

Embedded Systems Application Developer Intern

PCB CUPID, Bangalore, India

Jul 2025 – Present

Remote

- Developed embedded applications and IoT solutions using Glyph development boards (C2, C6, H2) with microcontroller programming, achieving precision control and hardware-software integration for production-ready systems
- Designed and prototyped embedded solutions using C/C++ programming, converting proof-of-concepts to scalable systems deployed in real-world industrial environments
- Implemented microcontroller firmware solutions with sensor integration, achieving seamless communication between hardware components and software algorithms for automated systems

Robotics Engineering Team Member and Technical Mentor

THINKBOTS Robotics Club, DCRUST

Oct 2023 – Present

Murthal, Haryana

- Engineered autonomous robotics systems including line-following robots and multi-terrain gripper mechanisms using Arduino microcontrollers, C++ programming, and sensor arrays, achieving 95% accuracy in path tracking algorithms
- Mentored 15+ junior developers on robotics fundamentals, Arduino programming, embedded systems design, and project management, improving team project completion rate by 40% through technical training programs
- Optimized robotics control systems through collaborative debugging, algorithm enhancement, and system integration, reducing development and troubleshooting time by 30% across multiple team projects

PROJECTS

XTorr - Multi-Threaded BitTorrent P2P Client — [GitHub](#) — *Java, Networking, Concurrency* Dec 2024

- Developed complete peer-to-peer BitTorrent client from scratch using Java, implementing multi-threaded architecture, TCP/IP socket programming, peer-wire protocol communication, tracker HTTP requests, and SHA-1 cryptographic validation for reliable file sharing
- Engineered high-performance concurrent download system using thread pools, ExecutorService, and synchronized data structures to fetch file pieces simultaneously from multiple peers, achieving 300% performance improvement over sequential downloads

Forge - PCB Design Visualization CLI Application — [GitHub](#) — *Python, SVG Processing* Aug 2025

- Created command-line interface application that automates conversion of KiCad SVG files into comprehensive PCB visualizations, implementing geometric mapping algorithms, coordinate transformation systems, and component placement automation with sub-millimeter precision
- Developed modular SVG processing pipeline with XML parsing, file I/O operations, and mathematical coordinate transformation algorithms, streamlining PCB design workflow for electrical engineers and reducing manual design time by 60%

Autonomous Navigation Line Following Robot — *Arduino Nano, C++, PID Control*

Mar 2025

- Built autonomous mobile robot with 8-IR sensor array, L298N motor driver, and custom PID control algorithm implementation, achieving 98% path accuracy through complex navigation scenarios including sharp corners and varying track conditions
- Optimized sensor calibration protocols, implemented digital filtering algorithms, and developed adaptive thresholding systems, reducing false positive readings by 85% and ensuring stable performance across diverse lighting conditions and surface materials

Dobosaurus - Interactive Game Development Project — GitHub — *Java, Swing GUI*

Nov 2024

- Developed arcade-style platformer game application using Java Swing framework, implementing responsive user input handling, procedural obstacle generation algorithms, collision detection systems, and real-time scoring mechanisms optimized for 60+ FPS performance
- Designed scalable object-oriented software architecture with separation of concerns, implementing MVC design patterns to separate game logic, graphics rendering, and user input systems for maintainable and extensible codebase

ACHIEVEMENTS & RECOGNITION

Smart India Hackathon - National Level Competition — *Team Leader & Software Developer*

2024

- Led development of AI-powered Face Recognition Surveillance System and Communication Platform for Missing Persons tracking at Simhastha Ujjain, implementing computer vision algorithms, database management, and real-time notification systems for Police Department

ADDITIONAL SKILLS

Operating Systems: Linux (Ubuntu, Arch Linux, CentOS), Windows, Unix

Development Tools: Git, GitHub, Vim, Debugging Tools

Soft Skills: Team Leadership, Project Management, Technical Documentation, Problem Solving